

VISHAL AGGARWAL

Python Developer | GenAI & LLM Solutions | Full-Stack AI Applications
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PROFESSIONAL SUMMARY

Python Developer with experience delivering client-facing AI and GenAI applications at Tata Consultancy Services, with a strong foundation in software design, code quality, and modular architecture. Skilled in Python (FastAPI, LangChain, LangGraph, Streamlit), REST API development, and end-to-end solution delivery — from requirements analysis and architecture review to deployment and user validation. Adept at translating complex technical requirements into scalable, maintainable systems. Eager to contribute to Infosys's digital transformation delivery practice as a Python Consultant.

TECHNICAL SKILLS

Python Stack	FastAPI, LangChain, LangGraph, Streamlit, TensorFlow, Scikit-Learn, PyPDFLoader, dotenv
AI / GenAI	RAG Pipelines, Prompt Engineering, Agentic AI Agents, FAISS Vector DB, HuggingFace Embeddings, Groq API (Llama 3.1)
Web & API	REST API Design, Node.js, JavaScript, HTML, CSS, Chart.js
Dev Practices	Code Review, Unit Testing, Modular Architecture, Solution Design, Technical Risk Planning
Tools	Git, Jira (Agile), Azure AI (learning), NLTK, Vectorization, CSV Analytics

PROFESSIONAL EXPERIENCE

Python Developer – AI/GenAI Solutions (System Engineer) | [Tata Consultancy Services \(TCS\)](#) 2025 – Present

Pune, Maharashtra

- Interfaced with internal stakeholders to gather requirements, validate solution architecture, and ensure high-quality deliverables for GenAI-powered applications built in Python.
- Designed and developed modular, scalable Python backends using FastAPI and LangChain, adhering to organizational coding standards and architecture guidelines.
- Led code quality assurance by implementing structured, maintainable codebases and reviewing solution designs end-to-end before deployment.
- Built autonomous GenAI agent workflows in Python (LangChain + LangGraph), integrating tool-use and prompt orchestration pipelines for intelligent task automation.
- Developed production-grade Streamlit interfaces for real-time AI interaction, reducing average user response wait time by 3 seconds through optimized pipeline design.
- Contributed to project estimation, effort planning, and technical risk identification for AI application delivery cycles.

KEY PROJECTS

RAG-Based Intelligent Document Assistant — *Python · LangChain · FastAPI · FAISS · Groq API* [\[TCS Internal\]](#)

- Designed and delivered a full Retrieval-Augmented Generation (RAG) system in Python using LangChain, FAISS vector indexing, and Groq Llama-3.1-8B to answer queries from client-uploaded PDFs with high contextual accuracy.
- Architected the semantic retrieval pipeline — from document ingestion (PyPDFLoader) and HuggingFace embedding generation to FAISS indexing and LLM response synthesis.
- Conducted internal design reviews and validated the system architecture prior to deployment, ensuring robustness and maintainability.

- Delivered an interactive Streamlit frontend with PDF upload, streaming response rendering, and error-handling, achieving a measurable 3-second reduction in user wait time.

ResumeEvaluator — *Agentic AI System · Python · LangGraph · FastAPI* [\[Individual Project\]](#)

- Architected a graph-based agentic AI evaluation system in Python using LangGraph and FastAPI, processing multi-dimensional inputs (skills, experience, education) through structured LLM-driven analysis.
- Designed and implemented semantic skill matching algorithms and structured scoring logic, translating complex NLP outputs into clear, recruiter-readable insights.
- Delivered a complete, full-stack solution: modular Python backend (FastAPI) and responsive HTML/CSS/JS frontend — demonstrating end-to-end Python solution delivery skills.

Conjecture.AI — *AI-Powered Data Analytics · Node.js · LLM Integration* [\[TCS AI Friday Hackathon\]](#)

- Built a serverless Node.js API integrating Groq LLaMA 3 to convert raw CSV datasets into human-readable analytical hypotheses, enabling rapid insight generation for non-technical stakeholders.
- Engineered a hybrid analytics pipeline combining statistical correlation/outlier detection with AI reasoning — validating technical requirements through empirical testing before delivery.
- Designed an interactive Chart.js visualization dashboard, translating complex AI outputs into intuitive, client-ready data stories.

Fake News Detection Model — *Deep Learning · Python · TensorFlow · NLP* [\[Individual Project\]](#)

- Developed and validated a TensorFlow deep learning classifier using NLTK preprocessing, TF-IDF vectorization, and Scikit-Learn, achieving reliable binary news classification.
- Applied systematic unit test planning and model evaluation to ensure output quality and reproducibility of results.

EDUCATION

B.E. Computer Science & Engineering | [Chandigarh University](#)

2020 – 2024

Kharar, Punjab | CGPA: 8.55 / 10

12th PCM (CBSE): 91.8% — Army Public School, Kamptee (2020) | 10th (CBSE): 94.2% — PKR Jain Sr. Secondary School, Ambala (2018)

WHY INFOSYS

- Hands-on experience in client-interfacing delivery environments — requirement analysis, architecture validation, and quality assurance on production Python systems.
- Strong code ownership mindset: structured Python codebases (FastAPI, LangChain), design review, and unit test planning built into every project.
- Experience with emerging AI/GenAI tooling (RAG, agentic workflows, LLMs) positions me to help Infosys clients navigate their AI-driven digital transformation.
- Consistent academic and professional track record demonstrating technical rigor, adaptability, and commitment to high-quality delivery.